

Computational Power of Hamiltonians in Quantum Computing

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Important research topics of quantum computer science

1. development of abstract **computational models** that determine the elementary transformations of a quantum computer
(based on physical laws governing the considered quantum system),
2. classification of **computational problems** with respect to their complexity within these models
3. development of **efficient quantum algorithms** to handle these various types of computational problems

development of *computational models* is central



complexity of problems and
efficiency of algorithms

Control theoretic model of quantum computers

respects more the properties of real quantum systems than the quantum circuit model

elementary operations of the control theoretic model:

- external control operations in control group K

unitary transformations V

- Hamiltonian time evolution internal to the system

$$\exp(-iHt)$$

Control theoretic model of quantum computers

implement U by sequences

perform V_1 , wait t_1 , perform V_2 , wait t_2 , $\dots$, perform V_N , wait t_N

such that

$$U = \exp(-iHt_N)V_N \cdots \exp(-iHt_2)V_2 \exp(-iHt_1)V_1$$

complexity of U

- number of control operations N and
- time $t_1 + t_2 + \dots + t_N$

Mutual simulation of time evolutions

use control theoretic model to implement

- transformations for solving computational problems
- transformations representing the time-evolution of another quantum system ($\rightarrow$ universal quantum simulator)

simulation cost defines “computational power” of different systems

determining the cost: difficult!

but: solvable for infinitesimal time intervals

Mutual simulation of **Hamiltonians**

mutual simulation of time evolutions for **infinitesimal** time intervals

simulated time evolution $\exp(-i \tilde{H} \epsilon)$ is reproduced with error $O(\epsilon^2)$

$$\tilde{H} = t_1 U_1^\dagger H U_1 + t_2 U_2^\dagger H U_2 + \dots + t_N U_N^\dagger H U_N$$



$$\exp(-i U_N^\dagger H U_N t_N \epsilon) \cdots \exp(-i U_2^\dagger H U_2 t_2 \epsilon) \exp(-i U_1^\dagger H U_1 t_1 \epsilon)$$

$$= \exp(-i \tilde{H} \epsilon) + O(\epsilon^2)$$

Mutual simulation of Hamiltonians

H can simulate $\tilde{H}$ with overhead 1, written $\tilde{H} \leq H$, and N time steps iff

$$\tilde{H} = \sum_{j=1}^N p_j U_j^\dagger H U_j ,$$

$p_1, \dots, p_N > 0$ and $p_1 + \dots p_N = 1$

$U_j \in K$ are control operations

$\tilde{H}$ is a convex sum of N unitary conjugates of H

H can simulate $\tilde{H}$ with overhead τ iff $\tilde{H} \leq \tau H$

time-optimal simulation problem $\longrightarrow$ convex optimization problem

Composite systems and pair-interaction Hamiltonians

system consisting of n qudits $\mathcal{H} = \mathbb{C}^d \otimes \mathbb{C}^d \otimes \cdots \otimes \mathbb{C}^d$

pair-interaction Hamiltonian

$$H = \sum_{k < l} \sum_{\alpha \beta} J_{kl; \alpha \beta} \sigma_{\alpha}^{(k)} \sigma_{\beta}^{(l)},$$

$B := \{\sigma_{\alpha} \mid \alpha = 1, \dots, d^2 - 1\}$ is a basis of $su(d)$

control operations are all local operations in

$$K := SU(d) \otimes SU(d) \otimes \cdots \otimes SU(d)$$

Main results of the thesis

- formal definition of mutual simulation of Hamiltonians
- lower and upper bounds on complexity of mutual simulation of Hamiltonians
- optimal simulation algorithms using methods of representation theory of finite groups (irreducible projective representations) design theory (orthogonal arrays, difference schemes, Hadamard matrices, and spreads) and algebraic graph theory
- application of mutual simulation of Hamiltonians to adiabatic quantum computing for solving computationally hard problems
- new computational problems and their quantum complexity classes

Algebraic Graph Theory gives Lower Bounds on Complexity

represent pair-interaction Hamiltonian

$$H = \sum_{k < l} \sum_{\alpha \beta} J_{kl; \alpha \beta} \sigma_{\alpha}^{(k)} \sigma_{\beta}^{(l)}$$

$B := \{\sigma_{\alpha} \mid \alpha = 1, \dots, d^2 - 1\}$ is a ONB of $su(d)$

as coupling matrix (extension of adjacency matrix of a graph)

$$J = \left(\begin{array}{c|c|c|c|c} 0 & J_{12} & J_{13} & \cdots & J_{1n} \\ \hline J_{21} & 0 & J_{23} & \cdots & J_{2n} \\ \hline \vdots & \vdots & & & \\ \hline J_{n1} & J_{n2} & J_{n3} & & 0 \end{array} \right) \in \mathbb{R}^{(d^2-1)n \times (d^2-1)n}$$

Analogy: coupling matrix $\leftrightarrow$ adjacency matrix

consider dipole-dipole coupling between all nodes

$$H = \sum_{k < l} w_{kl} (3 \sigma_z^{(k)} \sigma_z^{(l)} - \sum_{\alpha=x,y,z} \sigma_{\alpha}^{(k)} \sigma_{\alpha}^{(l)}) , \quad w_{kl} \neq 0$$

coupling matrix can be expressed as

$$J = \textcolor{red}{W} \otimes \textcolor{blue}{C} ,$$

where $W = (w_{kl})$ and $C = \text{diag}(-1, -1, 2)$

$\textcolor{red}{W}$ adjacency matrix of a weighted graph

$\textcolor{blue}{C}$ characterizes the type of coupling between nodes

Action of control operations on the coupling matrix J

Conjugation of the pair-interaction Hamiltonian H by control operations

corresponds to

conjugation of J by a direct sum of orthogonal matrices

$$H \mapsto (U_1 \otimes U_2 \otimes \cdots \otimes U_n)^\dagger H (U_1 \otimes U_2 \otimes \cdots \otimes U_n)$$

$$J \mapsto (O_1 \oplus O_2 \oplus \cdots \oplus O_n) J (O_1 \oplus O_2 \oplus \cdots \oplus O_n)^T$$

Lower bounds on time overhead

majorization criterion

H can simulate $\tilde{H}$ with time overhead τ

$$\sum_{j=1}^N p_j O_j (\tau J) O_j^T = \tilde{J}$$

$\tilde{J}$ convex sum of conjugates of $\tau J \implies \text{Spec}(\tilde{J})$ majorizes $\text{Spec}(\tau J)$

Lower bounds on time steps

rank criterion (special case)

H can simulate $\tilde{H}$ with N time steps

$$\tau \sum_{j=1}^N p_j O_j (W \otimes C) O_j^T = \tilde{W} \otimes C$$

adding the block diagonal matrix $R := \tau \sum_{j=1}^N p_j O_j (\mathbf{1} \otimes C) O_j^T$ on both sides gives

$$\implies N \geq \text{rank}(\tilde{W} \otimes C + R) / \text{rank}((W + \mathbf{1}) \otimes C)$$

Complexity of time reversal

time reversal: $H \mapsto -H$ (refocusing, spin echoes)

$J = W \otimes C$ coupling matrix of n coupled qubits, $w_{kl} \neq 0$ for all k, l

1. C traceless

time step and time overhead are at most 2

2. C has positive and negative eigenvalues, but $\text{tr}(C) \neq 0$

time steps at least $n/3 - 1$

time overhead independent of n , depends on eigenvalues only

3. C either positive or negative semidefinite

time steps at least $n - 1$

time overhead at least $n - 1$

Complexity of time reversal

dipole-dipole coupling between all nodes (easy to invert)

$$H = \sum_{k < l} (3 \sigma_z^{(k)} \sigma_z^{(l)} - \sum_{\alpha=x,y,z} \sigma_{\alpha}^{(k)} \sigma_{\alpha}^{(l)})$$

$$J = K \otimes \text{diag}(-1, -1, 2),$$

where K is the adjacency matrix of the complete graph

lower bound on time overhead is 2

this bound is tight

(well known-scheme in NMR with time overhead 2 and 2 time steps)

Complexity of time reversal

strong scalar coupling between all nodes (difficult to reverse)

$$H = \sum_{k < l} (\sigma_x^{(k)} \sigma_x^{(l)} + \sigma_y^{(k)} \sigma_y^{(l)} + \sigma_z^{(k)} \sigma_z^{(l)})$$
$$J = K \otimes \mathbf{1}_3$$

lower bound on the time overhead $n - 1$

→ reversed time-evolution necessarily slower by the factor $(n - 1)$

lower bound on time steps $n - 1$

upper bounds $cn - 1$

(decoupling algorithms based on orthogonal arrays)

Computational power of Hamiltonians

$$H_1 = \sum_{k < l} w_{kl} (\sigma_x^{(k)} \sigma_x^{(l)} + \sigma_y^{(k)} \sigma_y^{(l)})$$

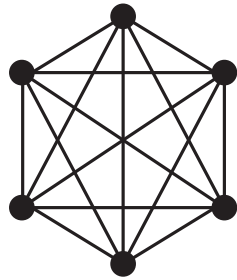
$$H_2 = \sum_{k < l} w_{kl} (\sigma_x^{(k)} \sigma_x^{(l)} - \sigma_y^{(k)} \sigma_y^{(l)})$$

H_1 cannot simulate H_2 with overhead less than $(n - 1)$

H_2 can simulate H_1 with overhead 2

Graph-theoretical bounds: Selective decoupling

complete graph

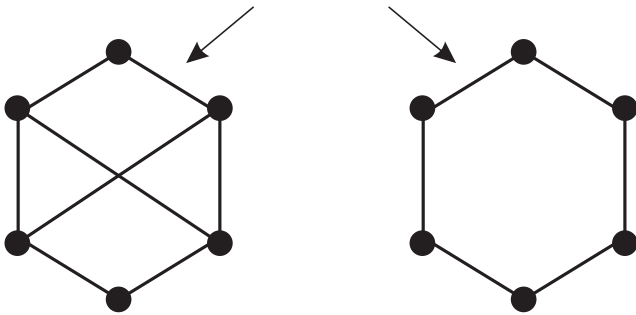


complete Ising-Hamiltonian:

$$\sigma_z \otimes \sigma_z \text{ between all qubits}$$

control operations **1** and σ_x

selective decoupling



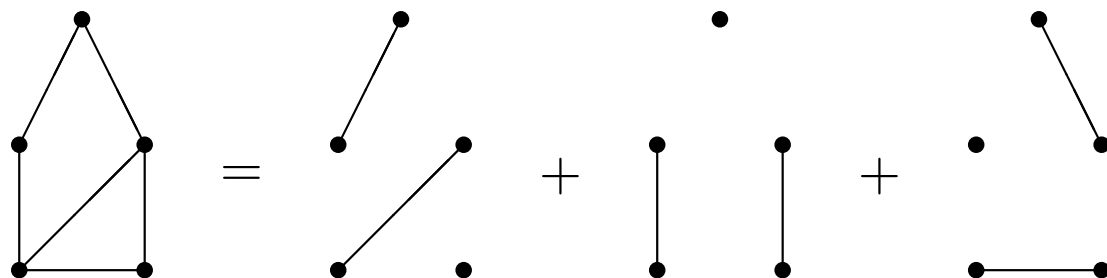
Combinatorial problem: switching off unwanted interactions

are the other interactions necessarily weakened (time overhead > 1)?

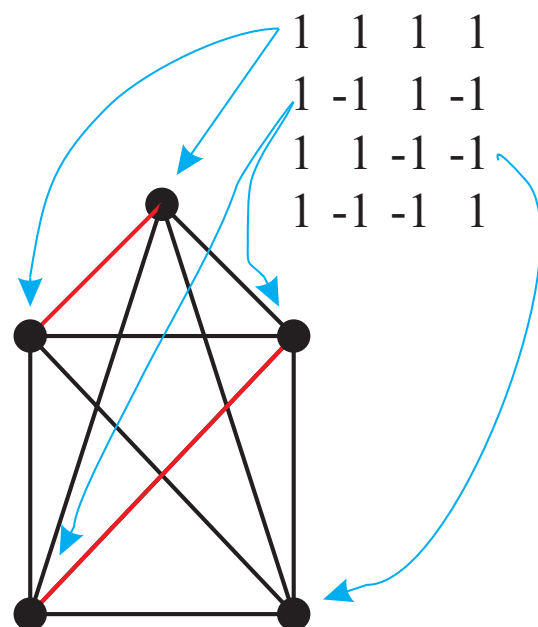
yes, for some graphs $\rightarrow$ lower bound on time overhead given by eigenvalues of the adjacency matrix

upper bound given by **chromatic index** of interaction graph

Selective decoupling based on chromatic index



Hadamard matrices: orthogonal matrices with ± 1 entries



1	1	1	1
1	-1	1	-1
1	1	-1	-1
1	-1	-1	1

+1: conjugation by $\mathbf{1}$

-1: conjugation by σ_x ($\sigma_x \sigma_z \sigma_x = -\sigma_z$)

rows: different pairs of qubits or single qubit

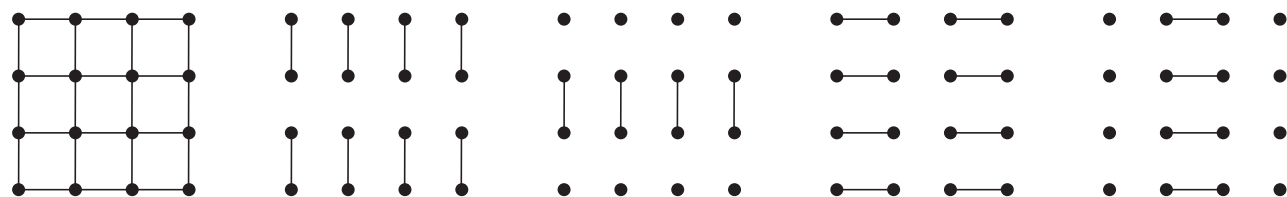
columns: time steps

Application: “one-way quantum computer”

required: Ising-interactions on two-dimensional lattice

but: resource Hamiltonian couples all qubits

→ selective decoupling



$l \times l$ lattice

selective decoupling with time overhead 4 (=chromatic index)

lower bound on time overhead

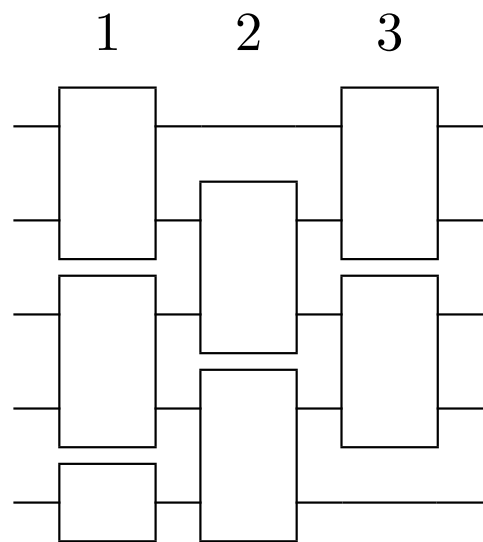
$$-4 \cos \left(\frac{\pi}{l+1} l \right) \approx 4$$

Application: continuous complexity measure for quantum circuits

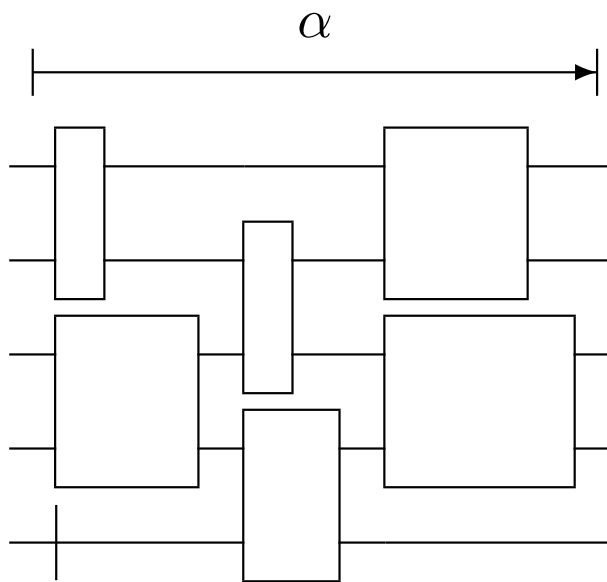
weighted depth:

continuous complexity measure

based on control theoretic model with complete Ising-Hamiltonian



a. depth



b. weighted depth

Application of mutual simulation of Hamiltonians to adiabatic quantum computing

To do: construct “physically reasonable” Hamiltonians whose ground states encode solutions of the problem “independent set”

- graph $G = (V, E)$, positive integer $v \leq |V|$
- $\exists$ independent set with cardinality at least v

subset $V' \subseteq V$ with

- $|V'| \geq v$ and
- no two vertices in V' are joined by an edge in E

“independent set” remains NP-complete for cubic planar graphs.

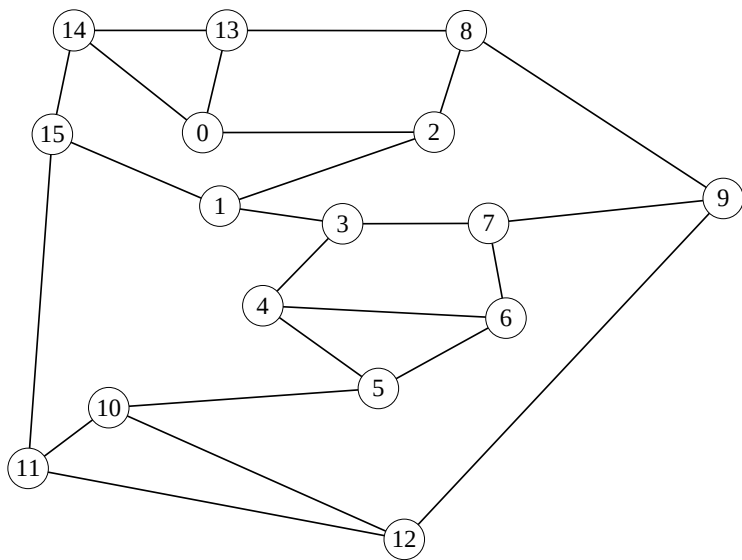
Theorem (Planar spin glass within a magnetic field)

$G = (V, E)$ cubic planar graph

determining energy of ground states of the Hamiltonian

$$\sum_{(k,l) \in E} \sigma_z^{(k)} \sigma_z^{(l)} + \sum_{k \in V} \sigma_z^{(k)}$$

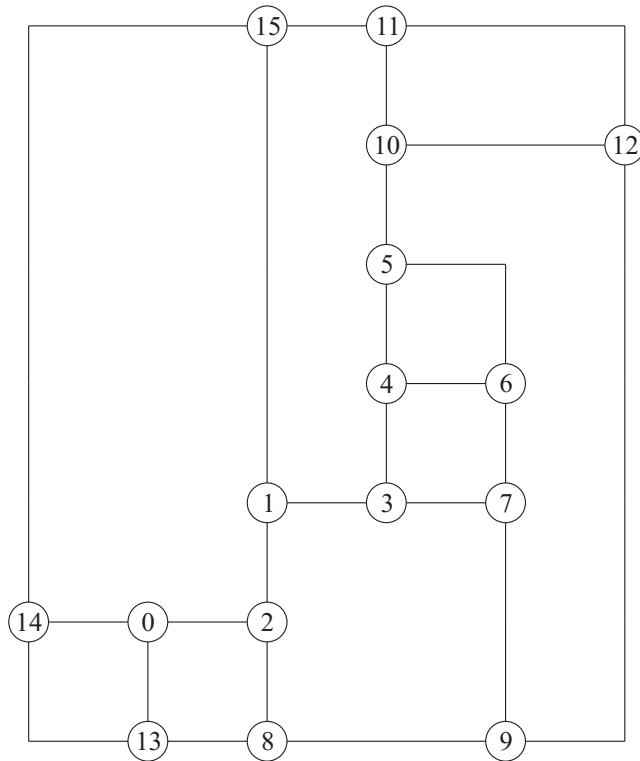
equivalent to determining maximal cardinality of independent sets



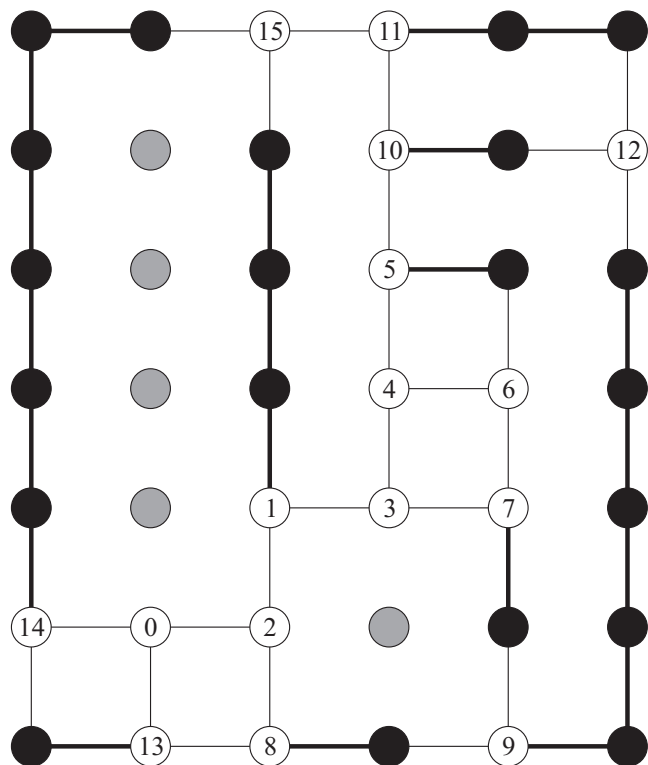
but not physically reasonable!

Planar orthogonal embedding of graph $G = (V, E)$ maps

- vertices $k \in V$ to lattice points $\Gamma(k)$
- edges $(k, l) \in E$ to paths such that
 - images of their endpoints $\Gamma(k)$ and $\Gamma(l)$ are connected and
 - paths do not intersect



Planar orthogonal Hamiltonians



filled circles	no σ_z
thick lines	$-c\sigma_z \otimes \sigma_z$
circles	σ_z
thin lines	$\sigma_z \otimes \sigma_z$

ground states correspond 1-to-1 to grounds states of H for $c = 3$

gap between ground and first excited is the same for $c = 9$

can be simulated efficiently by Ising-interactions on lattice

16 time steps, $2(c + 1)$ time overhead

Quantum complexity classes

determining minimal energy of

- pair-interaction (2-local) Hamiltonians: NP-complete
- k -local Hamiltonians: QMA-complete for $k \geq 3$

QMA/QCMA quantum extensions of NP

New complete problems for QMA/QCMA

low-energy and low-complexity states of local Hamiltonians

Non-identity check

Non-identity check on basis states

⇒ impression of how difficult it is to decide if an implementation in the control-theoretic model is optimal

→ justification for considering infinitesimal time intervals (mutual simulation of Hamiltonians)

Mathematical tools

Vector majorization

let $x = (x_1, x_2, \dots, x_d) \in \mathbb{R}^d$

define $x^\downarrow = (x_1^\downarrow, x_2^\downarrow, \dots, x_d^\downarrow) \in \mathbb{R}^d$ as the vector whose components are the same as of x , but ordered so that

$$x_1^\downarrow \geq x_2^\downarrow \geq \dots \geq x_d^\downarrow$$

x majorizes y , written $x \preceq y$, if

$$\sum_{j=1}^k x_j^\downarrow \leq \sum_{j=1}^k y_j^\downarrow$$

for $k = 1, \dots, d-1$ and

$$\sum_{j=1}^d x_j^\downarrow = \sum_{j=1}^d y_j^\downarrow$$

Mathematical tools

Operator majorization

let A and B be two Hermitian matrices of size d

we say A majorizes B , written $A \preceq B$, if

$$\lambda(A) \preceq \lambda(B),$$

where $\lambda(A)$ and $\lambda(B)$ are the spectra (the vectors of eigenvalues) of A and B , respectively.

Uhlmann's theorem

we have $A \preceq B$ iff

$$A = \sum_{j=1}^N p_j U_j^\dagger B U_j,$$

where $\{p_j\}$ is a probability distribution and $\{U_j\}$ is a set of unitaries